

Week 2 — Six Questions for Reading a Methods Section

SPPH 604 · Application of Advanced Epidemiological Methods

Running example: Kueh et al. (2026), MASLD / NHANES · previews L1

Six questions for reading a Methods section

Know what you are trying to rebuild, before you try to rebuild it.

- ▶ The six questions are the **spine**; the NHANES anatomy is what the answers are **made of**
- ▶ Everything named today — component files, variable names, the join key — you use on **Thursday** in L1, and again for **M0** (due Wk3)

Why the question comes before the data

- ▶ The instinct is “what did they run?” — a model family answers nothing on its own
- ▶ A **Cox model appears under several different research goals**
- ▶ Reading a Methods section *is* the first act of reproduction

By the end of today: this paper answers four of six questions clearly, one partially, and one not at all.

The six questions

1. What is the **research goal**?
2. Who is the **population**, and how is eligibility operationalised?
3. What is the **exposure / index factor**, and the comparison?
4. What is the **outcome**, time zero, and follow-up?
5. What **other variables** enter, and what role are they supposed to play?
6. What **design and analysis features** must be respected?
If any one is under-specified, the study is not reconstructable — no matter how clean the results look.

Q1 — What is the research goal?

The paper states two aims, not one

- ▶ **Aim 1** — the *clinical correlates* of four groups defined by BMI and waist-to-height ratio
- ▶ **Aim 2** — the *prognostic value* of that classification for mortality

	Clinical correlates	Mortality prognosis
Question	How do the four groups differ in health characteristics?	Do the four groups have different subsequent mortality?
Time structure	Cross-sectional at baseline	Longitudinal follow-up

A single paper can carry more than one research question, and they have different clocks.

Five goals the same variables can serve

Research goal	Scientific target in this example
Description	What are the characteristics and mortality experience of each phenotype group?
Association	Is phenotype statistically associated with subsequent mortality?
Prognostic factor	Does phenotype carry information over and above other known prognostic factors?
Prediction	How accurately can we predict an individual's risk, and does adding WHtR improve it?
Causal	What would mortality have been under a specified contrast that changed central adiposity?

*The variables can be identical. **The target quantity is not.***

Same variables, different analyses

Goal	Typical analysis	How are other variables handled?
Description	Weighted means, proportions; Kaplan–Meier or standardised survival	Stratification / standardisation where useful
Association	Cox regression, splines, group comparisons	Define a conditional association ; no automatic causal reading
Prognostic factor	Cox model holding established prognostic factors + the index factor	Adjust for other prognostic factors
Prediction	Develop/validate a survival model; base vs base + WHtR	Chosen for predictive value; shrinkage and validation
Causal	Target trial; standardisation, g-formula, IPW	Select confounders by causal knowledge; avoid mediators/colliders

*A Cox model appears in several rows. **The model family does not determine the goal.***

Same variables, different evidence of success

Goal	What would count as evidence
Description	Absolute summaries with uncertainty: means, prevalences, survival at meaningful times
Association	An association measure such as HR + 95% CI; assumptions and functional form
Prognostic factor	Adjusted effect; for <i>incremental</i> value, a model with vs without the factor
Prediction	Calibration, discrimination (C-index / time-dependent AUC), Brier score , validation
Causal	A prespecified contrast, positivity/balance diagnostics, sensitivity to unmeasured confounding

HR + CI: what it is, and is not, evidence for

HR + CI is enough to report an adjusted association, or a prognostic-factor effect.

It is **not** evidence that a prediction model is good, and **not** by itself evidence that a causal effect has been identified.

The paper names “prognostic value” and supplies adjusted association.

What does this paper most directly support?

It directly supports: description of how the groups differ; association between phenotype and subsequent mortality; an adjusted prognostic association.

It does not, by itself, establish: that central adiposity *causes* the difference; that reducing WHtR would reduce mortality; that adding WHtR improves individual prediction.

Q2 — Who is the population?

First: how NHANES is organised

- ▶ **Six two-year cycles**, 2007–2018, each with its own file suffix: E F G H I J
- ▶ Files are .XPT (SAS transport) — `haven::read_xpt()`, or `nhanesA::nhanes()`
- ▶ Each file is **one row per participant, for one topic** — not one big table
- ▶ Three collection settings: **Interview** (home) · **MEC exam** (measured body size, BP) · **Laboratory** (triglycerides, GGT, glucose)

Family	Examples	What it holds
Demographics	DEMO	age, sex, race, survey design, weights
Examination	BMX, BPX	measured body size, blood pressure
Laboratory	TRIGLY, BIOPRO, GLU, GHB, HDL	blood results
Questionnaire	ALQ, BPQ, DIQ, SMQ	self-reported history and medication

*Mortality is **not** an NHANES file. It is a separate linked, fixed-width file from NCHS.*

Where the measurements and labs come from

Concept	NHANES variable	Component
Participant ID	SEQN	<i>every file — the merge key</i>
Age, sex, race	RIDAGEYR, RIAGENDR, RIDRETH3	DEMO
BMI, waist, height	BMXBMI, BMXWAIST, BMXHT	BMX
Triglycerides	LBXTR	TRIGLY
GGT	LBXSGTSI	BIOPRO
Glucose, HbA1c	LBXGLU, LBXGH	GLU, GHB
HDL cholesterol	LBDHDD	HDL
Blood pressure	BPXSY1-4, BPXDI1-4	BPX

Anthropometry, blood pressure and laboratory values are **measured or assayed**. Age, sex and race/ethnicity come from the demographic file.

Where the questionnaire, design and outcome come from

Concept	NHANES variable	Component
BP / lipid medication	BPQ050A, BPQ100D	BPQ
Diabetes	DIQ010, DIQ050, DIQ070	DIQ
Alcohol	ALQ130, ALQ120Q, ALQ120U, ALQ121	ALQ
Survey design	WTMEC2YR, WTSAF2YR, SDMVSTRA, SDMVPSU	DEMO, TRIGLY
Mortality	MORTSTAT, PERMTH_EXM, ELIGSTAT	linked mortality file

*These variables alone reach across **seven examination and laboratory components**, plus demographics, plus questionnaire components, plus a separate linked mortality file. **That is what L1 assembles.***

SEQN is the join key

- ▶ Every respondent has one **SEQN** — the identifier CDC uses to sort and merge files
- ▶ Analysis = pick the topic files you need, **merge on SEQN**, one row per person
- ▶ SEQN does **not** repeat across the continuous cycles: all **59,842** values across 2007–2018 are distinct, and the per-cycle ranges do not overlap — *Analysis set: all NHANES 2007–2018 records, $N = 59,842$*
- ▶ The ranges are **not contiguous**, though — a gap of 1,640 sits between 2011–2012 and 2013–2014, so do not infer a cycle from an ID
- ▶ Keep a **cycle variable** anyway: provenance, definitions that change between cycles, and weights built per cycle. Not to disambiguate SEQN

Thursday, previewed: L1 builds exactly this

L1 in one screen: two files, one key

```
pacman::p_load(tidyverse, nhanesA, tableone, naniar)

demo <- nhanes("DEMO_H", translated = TRUE) # 2013-2014 demographics
bmx <- nhanes("BMX_H", translated = TRUE) # 2013-2014 body measures

merged <- full_join(demo, bmx, by = "SEQN")
```

- ▶ DEMO_H and BMX_H are the **same** DEMO and BMX from the “*Where the measurements and labs come from*” table, cycle **H**
- ▶ translated = TRUE turns stored codes into labels — convenient, and it changes what a value *looks* like, so recode rules written against raw codes will silently miss
- ▶ full_join **keeps people measured in one file but not the other**. That is a row-set decision made by a single word: inner_join would be a different study

Then do it for every cycle

```
cycles <- c("H", "I", "J") # 2013-14, 2015-16, 2017-18

all_cycles <- lapply(cycles, function(cy) {
  d <- nhanes(paste0("DEMO_", cy), translated = TRUE)
  b <- nhanes(paste0("BMX_", cy), translated = TRUE)
  full_join(d, b, by = "SEQN") %>% mutate(cycle = cy)
})

stacked <- bind_rows(all_cycles)
```

- ▶ `paste0("DEMO_", cy)` is why the **suffix table** was worth memorising
- ▶ `bind_rows()` stacks cycles; the joins happen **within** a cycle first
- ▶ Our reproduction does the same over **all six** cycles, E through J

Raw names in, analysis names out

```
analysis <- stacked %>%  
  rename(Sex = RIAGENDR, Age = RIDAGEYR,  
         RaceEthnicity = RIDRETH3, BMI = BMXBMI) %>%  
  filter(Age >= 20) %>%  
  mutate(BMICat = cut(BMI, c(0, 18.5, 25, 30, Inf), right = FALSE,  
                     labels = c("Underweight", "Normal weight",  
                                "Overweight", "Obese")))
```

- ▶ Every rename is a **decision you have to be able to defend**, not cosmetics
- ▶ **right = FALSE is not optional**. Without it cut() closes the intervals the other way and BMI exactly 30.0 lands in *Overweight* — one argument, and the categories are the mirror image of the ones you meant
- ▶ BMICat here uses the standard WHO cut-points. **The paper does not**: obesity at **BMI ≥ 30** , and **≥ 25** for Asian participants
- ▶ The lab keeps **age ≥ 20** ; the paper says **age > 18** ; our reproduction uses **≥ 18**
Same variable, three eligibility rules, three different studies.

Look at what is missing before you model it

```
colSums(is.na(analysis))  
naniar::gg_miss_var(analysis, show_pct = TRUE)  
  
tab1_vars <- c("Age", "RaceEthnicity", "BMI", "BMICat")  
CreateTableOne(vars = tab1_vars, data = analysis, strata = "Sex",  
               addOverall = TRUE, test = FALSE)
```

- ▶ Missingness is **inspected first**, then decided about — never discovered inside a model
- ▶ CreateTableOne is the Table 1 you will rebuild for the paper's cohort in **M1**
- ▶ test = FALSE on purpose: a p-value comparing baseline groups answers no question anyone asked

BREAK

Q2, continued — the population this paper defines

Population: who is being studied?

Adults with MASLD in NHANES, 2007–2018

MASLD = metabolic dysfunction-associated steatotic liver disease: excess liver fat together with metabolic dysfunction.

*But nobody looked at a liver. **Steatosis was estimated by a formula.***

The study population is **not the general population**. Everyone in the main analysis already meets the study's own definition of MASLD.

How MASLD was actually ascertained

Hepatic steatosis was **estimated** with the **Fatty Liver Index** (Bedogni 2006), thresholded at **FLI $\geq$ 60**:

FLI input	NHANES variable	Component
Triglycerides	LBXTR	TRIGLY
BMI	BMXBMI	BMX
GGT	LBXSGTSI	BIOPRO
Waist circumference	BMXWAIST	BMX

The four enter a published logistic formula — **not a product of the four**. What matters here is that the index is *computed*, from four variables in three component files.

Plus **at least one of five cardiometabolic criteria**: adiposity; raised glucose/HbA1c or diabetes; raised blood pressure or treatment; raised triglycerides or treatment; low HDL.

So “MASLD” here is a **derived variable, not a diagnosis** — no imaging, no biopsy.

The funnel, with a type on every row

Step	N	Kind of step
NHANES 2007–2018 records	59,842	—
Age ≥ 18	36,580	eligibility — drops 23,262
In fasting subsample	14,962	design — drops 21,618
FLI computable	14,170	missingness
Steatosis (FLI ≥ 60)	6,052	target population — drops 8,118
MASLD (≥ 1 criterion)	6,050	target population
Mortality-linked	6,048	outcome ascertainment

After eligibility, the two largest drops are **design** and **disease definition** — and neither is missing data.

Analysis set: locked mortality-linked cohort, $N = 6,048$.

What the funnel reveals

Reproducibility note. The published paper reports ~**6,300** participants with MASLD; this pipeline's full reconstructed analytic file has **6,371**. The funnel's **6,048** is a further subset, those with a valid fasting weight. *Analysis set: full reconstructed analytic file, $N = 6,371$; locked cohort, $N = 6,048$.*

The paper's ~6,300 and our 6,371 should not be presented as if they described the same cohort.

The paper also states eligibility as **age** > 18 , while this reproduction uses **age** ≥ 18 . Small implementation differences like this are **recorded, not silently harmonised**.

Q3 — Exposure and comparison

The exposure is constructed, not measured

The exposure is not one measurement. It is built in two steps:

$BMXWAIST \div BMXHT \rightarrow$ **waist-to-height ratio** $\cdot BMXBMI \rightarrow$ **obesity category**

$\rightarrow$ BMI category $\times$ WHtR category $\rightarrow$ **four phenotype groups**

- ▶ All three inputs come from **one component**, BMX, in the MEC exam — so a person missing the exam is missing the exposure entirely
- ▶ Contrast that with **eligibility**, which needs BMX **and** the laboratory (LBXTR, LBXSGTSI) — eligibility spans two settings, the exposure does not

Exposure component 1: BMI

$$\text{BMI} = \text{weight (kg)} / \text{height}^2 (\text{m}^2)$$

- ▶ Weight and height were **measured** in the exam, not self-reported (BMXBMI)
- ▶ BMI marks **overall body size** — not body fat, and not where fat is stored
- ▶ Obesity defined as **BMI ≥ 30** ; for Asian participants **BMI ≥ 25** (RIDRETH3 == 6)
***Paper reporting note.** The Methods define obesity as BMI ≥ 30 (≥ 25 for Asian participants), but published Tables 1–2 label the groups as BMI ≥ 25 versus < 25 . Those labels conflict with the Methods and with the reported group BMI distributions, so they appear to be a reporting error.*

Exposure component 2: waist-to-height ratio

$$\text{WHtR} = \text{waist (cm)} / \text{height (cm)} = \text{BMXWAIST} / \text{BMXHT}$$

A person 170 cm tall with a 102 cm waist has $\text{WHtR} = 102/170 = \mathbf{0.60}$.

- ▶ High central adiposity is defined as **$\text{WHtR} \geq 0.60$**
- ▶ WHtR is a simple marker of **abdominal adiposity** — not CT or MRI visceral fat

The exposure is the combination

	Lower central adiposity — WHtR < 0.60	High central adiposity — WHtR $\geq$ 0.60
Obesity by BMI	Group I — <i>reference</i>	Group II
No obesity by BMI	Group III	Group IV

Group IV — *not obese by BMI, but a large waist for their height. The study asks whether BMI alone misses this higher-risk phenotype.*

- ▶ There is no treated arm and control arm: the **four-category phenotype is the index factor**, and mortality is compared **across** the categories
- ▶ Every hazard ratio is measured **against Group I**. A ratio without its referent means nothing — and here the referent is an **obese** group

Eligibility and exposure share their inputs

	Inputs
Eligibility ($FLI \geq 60$)	triglycerides, GGT, BMI , waist
Exposure (phenotype group)	BMI , waist $\div$ height

*Being selected into the study and being placed in a phenotype group are **not independent events**. That is **structural dependence**.*

For a causal question, conditioning on this selected population could open selection or collider paths under some structures; for a prognostic question, it primarily defines a selected target population. It is not automatically circularity, and not automatically bias.

Never trust a label you have not recomputed.

Q4 — Outcome, time zero, follow-up

Outcome and the clock

- ▶ **All-cause mortality** (MORTSTAT) via the NCHS linked mortality file — *not* an NHANES file, a separate linked fixed-width one
- ▶ **Time zero is the MEC examination**; PERMTH_EXM is person-months from that exam
- ▶ Follow-up through **31 December 2019**, stated in the NCHS 2019 Linked Mortality File linkage methodology document — not inferred from the filename
- ▶ Participants entered between 2007 and 2018, so **follow-up lengths and censoring differ** — time-to-event methods are appropriate. A **risk difference at a fixed horizon can still be estimated**, using methods that account for censoring

Time zero is a question the checklist never asks

- ▶ “Exposure, outcome, covariates” can all be specified while **time zero stays ambiguous**
- ▶ An ambiguous time zero silently changes **who is at risk, and for how long**
- ▶ Here exposure is measured **at** time zero — convenient, and rarer than you would think

Ask of your M0 candidates: when does follow-up begin, and does everyone start together?

Q5 — Other variables, and their roles

“In the model” is not the same as “confounding”

- ▶ The paper adjusts for age, sex, comorbidities, medications
- ▶ **“We adjusted for X” says where X sits in the syntax. It does not say what X does.**
- ▶ The same variable can be a confounder, a mediator, a competing prognostic factor, or a collider — and which it is depends on the **research goal from Q1**

What a crude-to-adjusted change establishes

- ▶ Group IV vs I all-cause HR $\approx$ **15.1 crude** $\rightarrow \approx$ **2.9 with the paper's Model 2**
- ▶ Conditioning on the adjustment set moves the estimate **enormously**
- ▶ What that movement *means* — confounding controlled, mediation removed, a collider opened — depends on the goal and the causal role of those variables

Analysis set: full reconstructed analytic file, $N = 6,371$ (Model 2 on its complete cases); unweighted, as in the paper.

Five NHANES traps — each has bitten this project

- ▶ **ALQ130 is drinks per *drinking day*, not drinks per day.** A “>2 a day” rule applied to it directly removes **2,063** people here, against **427** for average intake (frequency $\times$ quantity): about **1,600** over-excluded. **Week 3 owns this one**
- ▶ **777 = refused, 999 = don't know — codes, not counts.** Left inside a threshold rule they reclassified **nine** such records as heavy drinkers. Two traps on one variable, a factor of about two hundred apart
- ▶ **RIDRETH3 does not exist before 2011.** It is what identifies Asian participants for the lower BMI threshold — so the first two cycles cannot apply the paper's own rule
- ▶ **Fasting labs carry their own weight.** LBXTR and LBXGLU come from the fasting subsample, weighted by WTSAF2YR, not WTMEC2YR
- ▶ **Blood pressure is not a row mean** — next slide

Analysis set: full reconstructed analytic file, $N = 6,371$ (the alcohol counts).

The blood pressure trap, in full

The NHANES **Blood Pressure Procedures Manual (s3.1.7.3)** specifies the averaging:

- ▶ The **first reading is excluded** whenever more than one was obtained
- ▶ An **all-zero diastolic set averages to zero** — a diastolic 0 is a *measurement*, and NHANES codes a failure to obtain a reading separately, as missing
- ▶ A **zero sitting beside a positive reading is omitted** from the average

Applying it changed the derived diastolic pressure for **5,523 of 6,371** participants and moved hypertension from 3,444 to 3,448 — while the Model 1 and Model 2 hazard ratios shifted only in the third significant figure. *Analysis set: full reconstructed analytic file, N = 6,371.*

A coding error can be real, and worth fixing, even when the headline estimate survives it.

Q6 — Design and analysis features

NHANES is not a simple random sample

- ▶ A complex multistage probability design: **weights**, **strata** (SDMVSTRA), **PSUs** (SDMVPSU)
- ▶ We *use* them in **Week 6**; today, note they exist and that ignoring them is a choice
- ▶ Fasting-subsample analyses carry their own weight — WTSAF2YR, not WTMEC2YR
- ▶ Build the design on the **full frame**, then `subset()` — never on the subsample alone

Designed subsampling is not missingness

- ▶ Sampling into the fasting subsample is **statistically designed**, with a weight built to keep it nationally representative; compliance and lab nonresponse also play a part
- ▶ Check rather than assume: adults in and out of it differ by **0.2 years of age, 0.02 kg/m² BMI, 0.07 cm waist**
- ▶ It is balanced on the very measurements this paper's exposure is built from

Analysis set: all NHANES adults 2007–2018, $N = 36,580$.

Designed subsampling $\neq$ ordinary missingness.

Survey design: paper versus reproduction

- ▶ The published paper's **Statistical analysis section does not report** using NHANES survey weights, strata or PSUs
- ▶ Our pipeline **does** include design variables and the fasting-subsample weight
- ▶ Therefore a survey-weighted rebuild is a **design-informed reanalysis**, not a literal reproduction of every published analytic choice
Without the design information, estimates and standard errors should not be described as design-based nationally representative estimates.

Closing

PECOT, and the question rebuilt

Element	This study
P — Population	Adults with MASLD by $FLI \geq 60 + \geq 1$ cardiometabolic criterion, in the fasting subsample
E — Exposure	Four-category BMI $\times$ WHtR phenotype
C — Comparison	Groups II, III, IV versus Group I
O — Outcome	All-cause mortality to 31 Dec 2019
T — Time	Months from exam to death or censoring

Among adults with MASLD, are BMI/WHtR phenotypes associated with different subsequent risks of all-cause mortality?

M0 — ask the six of your own papers

- ▶ Shortlist two or three candidate NHANES papers; answer **all six**, each with a pointer to where in the paper you found it
- ▶ **Reconstructable is necessary, not sufficient** — a perfectly specified paper can still be a poor M0 choice
- ▶ M0 wants: enough observations and events, a design feature its analysis must respect, and a missing variable the analysis actually uses
- ▶ An unanswered question is not a disqualification — it is a candidate for the judgement you defend at **M1**

Bridge to L1 and M0

- ▶ **Thursday (L1):** the code you saw before the break — several cycles downloaded, topic files read, merged on SEQN, renamed, categorised, missingness inspected, Table 1 built
- ▶ **Before you go:** you should be able to name the component file each variable on these slides comes from
- ▶ **Toward M0 (Wk3):** bring six answers per candidate paper, not a summary

Reading a Methods section is the first act of reproduction.

Reference

Kueh MTW, Intaran MAU, Goh R, et al. **Body weight categories and fat distribution in relation to all-cause mortality among adults with metabolic dysfunction-associated steatotic liver disease: a population-based analysis of NHANES, 2007–2018.** *BMJ Open*. 2026;16:e113719. doi: 10.1136/bmjopen-2025-113719

Methods references: Riley RD et al. *BMJ* 2019;364:k4597 (prognostic factors) · Moons KGM et al. *BMJ* 2009;338:b375 and Riley RD et al. *BMJ* 2024;386:e078276 (prediction models) · Hernán MA, Robins JM. *JECH* 2006 and Hernán MA, Dahabreh IJ. *Ann Intern Med* 2025 (causal estimands).

Cohort counts and variable names on these slides come from this repository's own reproduction pipeline, not from the paper's prose.